

Zin Hmue Paing

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EDUCATION

Temasek Polytechnic

Singapore

Diploma in Computer Engineering; CGPA: 4.0/4.0 (Ranked #1, Common Engineering Programme) Apr 2024 – May 2027

- **Coursework:** Python Programming (D), OOP / Java (D), Full-Stack Development (D), AI & Machine Learning (D)
D = Distinction
- **Awards:** Engineering Scholarship (3× recipient), Director's List (Top 10%, 2024/25 & 2025/26)

EXPERIENCE

IT Intern (Data Platform Engineering)

May 2026 – Present

The Coca-Cola Company (Pacific Refreshments Pte. Ltd.)

Singapore

- Owned end-to-end development of a composite Power BI semantic model integrating live DirectQuery sources (Dynamics 365, GARP) with SharePoint data, replacing manual spreadsheet reconciliation and giving three global regions a single real-time view of incident, request, and asset status for 1,200+ devices
- Identified a DAX logic error inflating reported MTTR by summing instead of averaging resolution times; corrected six measures to restore accuracy of KPIs reviewed weekly by IT leadership
- Reverse-engineered undocumented GARP business logic to deliver lifecycle reporting for 1,200+ enterprise devices, eliminating reliance on legacy dashboards and reducing manual asset-refresh tracking

Software Engineer Intern

Apr 2026 – May 2026

Datality Lab / Moodie.AI

Singapore

- Developed a zero-backend job skills portal converting a 272K-row SkillsFuture dataset into an in-browser SQLite database (sql.js/WebAssembly), delivering sub-100 ms search across 50K+ skill-to-role mappings
- Designed a speech analytics pipeline that processed 17 hours of Cantonese smartwatch audio from 5 ASD students, automatically flagging ~3,950 pitch/volume anomalies using adaptive per-student baselines and replacing manual therapist review of raw recordings
- Parallelized audio processing across multiple CPU cores, reducing analysis time by ~50% through 15-minute chunking; implemented longitudinal trend detection to surface behavioral drift across 25 sessions

PROJECTS

Lizzie – AI Desktop Automation Agent | *Electron, Node.js, Win32 UIA, Whisper ONNX, PowerShell*

Jun 2026

- Developed a Windows AI overlay that automates 13 desktop interactions through Win32 accessibility APIs, completing common workflows in ~2 seconds per step with fully offline voice input via a bundled Whisper ONNX model
- Optimized a batched UIA automation engine with DLL caching that eliminated 700 ms cold-start penalties per step, enabling reliable multi-step orchestration across Win32, UWP, and browser targets
- Unified Claude, Gemini, and GPT behind a provider-agnostic HTTP contract; shipped a one-click installer with DPAPI-encrypted credential storage and zero external dependencies for voice; validated with 8 users across student and non-technical segments

KakiLearn AI | *Next.js 16, React 19, TypeScript, Tailwind CSS v4, Claude Haiku, Google Cloud TTS*

Feb 2026

- Shipped a mobile-first learning platform helping seniors practice digital banking through interactive simulators (DBS PayNow, AR QR payments), voice-graded quizzes, and AI-generated courses in 4 languages
- Designed a voice companion state machine using browser-native STT and streaming TTS; created a course generation pipeline that converts spoken topics into structured lessons via Claude and YouTube Data API v3

Career Quest Map | *Python, Pygame, LangChain, Azure OpenAI* | Finalist, Top 4/40 teams, NTU Hackathon

Jan 2026

- Led development of an AI career exploration tool using structured prompting, role-play, and decision trees; achieved 4.4/5.0 user satisfaction and 100% completion rate across 13 sessions with 11 students

TECHNICAL SKILLS

Languages: Python, Java, SQL, HTML/CSS

Frameworks: React, Next.js, Flask, Tailwind CSS, Electron, TensorFlow, scikit-learn, LangChain

AI & Cloud: Claude API, OpenAI API, Gemini API, Whisper ONNX, Google Cloud TTS, Azure, GCP, Vercel

Developer Tools: Git, Power BI (DAX/DirectQuery), SQLite, MQTT, InfluxDB, Grafana, Win32 UIA

LEADERSHIP & CERTIFICATIONS

President, Computer Engineering Network (CENT), Temasek Polytechnic

Apr 2025 – Apr 2026

Peer Tutor, Digital Fundamentals & Engineering Mathematics, School of Engineering

Apr 2025 – Feb 2026

Certs: Microsoft Azure AI Apps and Agents Developer Associate | Google AI Professional | NVIDIA Deep Learning | CS50x